

Cumulative Final Review: A Geometric Decision Framework

Thursday, December 10, 2026

Visualization: State space, diagnostic data, and global method

system	diagnostic data	global method
$\dot{x} = f(x)$	$f(x_*) = 0$, $\text{sign } f$	phase line and uniqueness
$\dot{\theta} = f(\theta)$ on S^1	$f(\theta_*) = 0$, $\text{sign } f$	$f \neq 0$: $T = \left \int_0^{2\pi} \frac{d\theta}{f(\theta)} \right $
$\dot{z} = Az$ $\dot{z} = F(z)$ in $\mathbb{R}^2$	$\text{tr } A$, $\det A$, $(\text{tr } A)^2 - 4 \det A$ $F(z_*) = 0$, $DF(z_*)$	eigendirections and exact linear flow invariance, energy, symmetry, global theorems

Observation: Hypothesis for the circle-period formula

The circle-period formula applies when $f(\theta) \neq 0$ for every $\theta \in S^1$; continuity then forces f to have one sign on S^1 .

Observation: Scope of nonlinear linearization

For nonlinear planar systems, a hyperbolic Jacobian gives only the local portrait near the equilibrium.

Recall: Local conclusion

A **local conclusion** holds on a neighborhood of a specified state.

Recall: Global conclusion

A **global conclusion** holds on an explicitly stated entire domain.

Recall: Necessary condition

A condition is **necessary** for a conclusion if the conclusion implies the condition.

Recall: Sufficient condition

A condition is **sufficient** for a conclusion if the condition implies the conclusion.

Question: A scalar saddle-node

Analyze

$$\dot{x} = \lambda + x^2$$

for every real λ .

Solution. If $\lambda < 0$, the equilibria are

$$x_- = -\sqrt{-\lambda}, \quad x_+ = \sqrt{-\lambda}.$$

Since $f'(x) = 2x$, the equilibrium x_- is asymptotically stable and x_+ is unstable. The phase line is

x	$(-\infty, x_-)$	x_-	(x_-, x_+)	x_+	(x_+, ∞)
$\dot{x}$	+	0	-	0	+

At $\lambda = 0$, the origin attracts from the left and repels from the right:

x	$(-\infty, 0)$	0	$(0, \infty)$
$\dot{x}$	+	0	+

Thus the origin is half-stable and Lyapunov unstable. If $\lambda > 0$, there is no equilibrium and $\dot{x} > 0$ everywhere.

Question: Saddle-node bottleneck time

Let $A > 0$ and $\lambda > 0$. For

$$\dot{x} = \lambda + x^2,$$

compute the travel time from $x = -A$ to $x = A$ and its leading behavior as $\lambda \downarrow 0$.

Solution. The travel time is

$$\begin{aligned} T_A(\lambda) &= \int_{-A}^A \frac{dx}{\lambda + x^2} \\ &= \frac{2}{\sqrt{\lambda}} \arctan\left(\frac{A}{\sqrt{\lambda}}\right) \sim \frac{\pi}{\sqrt{\lambda}} \quad (\lambda \downarrow 0). \end{aligned}$$

Question: The saddle-node on a circle

Let $a > 0$. Analyze

$$\dot{\theta} = \omega - a \sin \theta, \quad \theta \in S^1,$$

for every $\omega \in \mathbb{R}$.

Solution. If $|\omega| < a$, define

$$\alpha = \arcsin(\omega/a), \quad -\frac{\pi}{2} < \alpha < \frac{\pi}{2}.$$

The equilibria modulo 2π are

$$\theta_s = \alpha, \quad \theta_u = \pi - \alpha.$$

Since

$$f'(\theta) = -a \cos \theta,$$

θ_s is asymptotically stable and θ_u is unstable. At $|\omega| = a$, they coalesce in a saddle-node. At $\omega = a$, the fixed point $\theta = \pi/2$ attracts from the left and repels from the right. At $\omega = -a$, the fixed point $\theta = 3\pi/2$ attracts from the right and repels from the left. In either threshold case the fixed point is half-stable and Lyapunov unstable. If $|\omega| > a$, there is no equilibrium and the phase rotates with the sign of ω .

Question: Rotation period beyond the circle saddle-node

Let $a > 0$ and $|\omega| > a$. Compute the positive rotation period of

$$\dot{\theta} = \omega - a \sin \theta, \quad \theta \in S^1.$$

Solution. For $\omega > a$, set $\phi = \theta - \pi/2$ and $u = \tan(\phi/2)$. Then

$$\begin{aligned} T &= \int_0^{2\pi} \frac{d\theta}{\omega - a \sin \theta} \\ &= \int_{-\pi/2}^{3\pi/2} \frac{d\phi}{\omega - a \cos \phi} \\ &= \int_{-\pi}^{\pi} \frac{d\phi}{\omega - a \cos \phi} \quad \text{by } 2\pi\text{-periodicity} \\ &= \int_{-\infty}^{\infty} \frac{2 du}{(\omega - a) + (\omega + a)u^2} \\ &= \frac{2\pi}{\sqrt{\omega^2 - a^2}}. \end{aligned}$$

For $\omega < -a$, the orientation reverses and the positive period has the same formula.

Recall: Trace–determinant notation

For a real 2×2 matrix A , write

$$\tau = \operatorname{tr} A, \quad \Delta = \det A, \quad D = \tau^2 - 4\Delta.$$

Visualization: Trace–determinant classification

For the linear system $\dot{z} = Az$,

condition	linear type
$\Delta < 0$	saddle
$\Delta > 0, D > 0$	node; stability from $\operatorname{sign} \tau$
$\Delta > 0, D < 0, \tau \neq 0$	spiral; stability from $\operatorname{sign} \tau$
$\Delta > 0, D < 0, \tau = 0$	center
$D = 0, \Delta > 0$	star or degenerate node; check eigenvectors
$\Delta = 0$	non-isolated equilibrium set

Negative trace means asymptotically stable; positive trace means unstable whenever $\Delta > 0$ and the equilibrium is hyperbolic.

Question: The same Jacobian can hide different nonlinear behavior

Compare

$$\dot{x} = y, \quad \dot{y} = -x$$

with

$$\dot{x} = y, \quad \dot{y} = -x - y(x^2 + y^2).$$

Solution. Both systems have

$$DF(0,0) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

and eigenvalues $\pm i$. For the first system,

$$E(x,y) = \frac{1}{2}(x^2 + y^2)$$

satisfies $\dot{E} = 0$, so its positive level sets are periodic orbits and the origin is a center.

For the second system,

$$\dot{E} = x\dot{x} + y\dot{y} = -y^2(x^2 + y^2) \leq 0.$$

Every disk

$$D_c = \{(x,y) : E(x,y) \leq c\} \quad (c \geq 0)$$

is compact and positively invariant. A nonconstant periodic solution of period T would satisfy

$$0 = E(x(T), y(T)) - E(x(0), y(0)) = - \int_0^T y(t)^2 (x(t)^2 + y(t)^2) dt.$$

The integrand would vanish identically. If the solution were nonzero at some time, then $x^2 + y^2 > 0$ on a time interval, so $y = 0$ on that interval. The equations would then give

$$0 = \dot{y} = -x,$$

Question: The same Jacobian can hide different nonlinear behavior (continued)

a contradiction. Hence $x \equiv y \equiv 0$, and there is no nonconstant periodic orbit.

Every positive semi-orbit remains in the compact disk $D_{E(x(0), y(0))}$. Its omega-limit set must contain an equilibrium; otherwise the Poincaré–Bendixson theorem would make it a periodic orbit. The origin is the only equilibrium, so $(0, 0)$ belongs to the omega-limit set. Since $E(x(t), y(t))$ is nonincreasing, it has a limit; a subsequence converging to $(0, 0)$ shows that this limit is 0. Hence

$$(x(t), y(t)) \longrightarrow (0, 0) \quad (t \rightarrow \infty).$$

Also,

$$E(x(t), y(t)) \leq E(x(0), y(0))$$

proves Lyapunov stability. Thus the origin of the nonlinear system is globally asymptotically stable, not a center.

Question: A parameter creates a unique stable limit cycle

Let $\mu \in \mathbb{R}$. Analyze

$$\dot{x} = \mu x - y - (x^2 + y^2)x,$$

$$\dot{y} = x + \mu y - (x^2 + y^2)y.$$

Solution. At the origin,

$$DF(0, 0) = \begin{pmatrix} \mu & -1 \\ 1 & \mu \end{pmatrix}, \quad \lambda = \mu \pm i.$$

For $r > 0$,

$$\begin{aligned} r\dot{r} &= x\dot{x} + y\dot{y} \\ &= \mu r^2 - r^4, \end{aligned}$$

so

$$\dot{r} = r(\mu - r^2).$$

Also,

$$\dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2} = 1.$$

Therefore

$$\begin{array}{l|l} \mu < 0 & \dot{r} < 0 \text{ for } r > 0, \quad r(t) \rightarrow 0 \\ \mu = 0 & \dot{r} = -r^3 < 0 \text{ for } r > 0, \quad r(t) \rightarrow 0 \\ \mu > 0 & \dot{r} > 0 \text{ for } 0 < r < \sqrt{\mu}, \quad \dot{r} < 0 \text{ for } r > \sqrt{\mu} \end{array}$$

Thus the origin is globally asymptotically stable for $\mu \leq 0$ and unstable for $\mu > 0$. When $\mu > 0$, the invariant circle $r = \sqrt{\mu}$ attracts every solution with $r(0) > 0$ and has period 2π . On every other positive-radius interval, $r(t)$ is strictly monotone, so no other periodic orbit exists. Thus $r = \sqrt{\mu}$ is the unique stable limit cycle.

Visualization: Criteria for periodic-orbit questions

criterion	conclusion
$\dot{V} < 0$ away from equilibria	no periodic orbit
$\nabla \cdot (BF)$ has one strict sign on a simply connected domain	no periodic orbit in that domain
compact trapping region with no equilibrium	at least one periodic orbit
isolated closed orbit	limit cycle
all Liénard hypotheses	exactly one stable limit cycle

Question: A scaled Liénard oscillator

Let $a > 0$ and $\mu > 0$. Prove that

$$\ddot{x} + \mu(x^2 - a^2)\dot{x} + x = 0$$

has exactly one stable limit cycle.

Solution. For Liénard's theorem,

$$f(x) = \mu(x^2 - a^2), \quad g(x) = x,$$

and

$$F(x) = \int_0^x f(u) du = \mu \left(\frac{x^3}{3} - a^2 x \right).$$

The function g is odd and positive for $x > 0$, and f is even. The only positive zero of F is

$$x = \sqrt{3}a.$$

Moreover,

$$F(x) < 0 \quad (0 < x < \sqrt{3}a),$$

and for $x > \sqrt{3}a$,

$$F(x) > 0, \quad F'(x) = \mu(x^2 - a^2) > 0, \quad F(x) \rightarrow \infty \quad (x \rightarrow \infty).$$

Thus there is exactly one stable limit cycle for every $\mu > 0$.

Question: Period of the scaled relaxation oscillator

Let $a > 0$. Estimate the period of the stable limit cycle of

$$\ddot{x} + \mu(x^2 - a^2)\dot{x} + x = 0$$

as $\mu \rightarrow \infty$.

Solution. Set

$$F_0(x) = \frac{x^3}{3} - a^2 x, \quad w = \dot{x} + \mu F_0(x), \quad y = \frac{w}{\mu}.$$

Then

$$\dot{x} = \mu(y - F_0(x)), \quad \dot{y} = -\frac{x}{\mu}.$$

Question: Period of the scaled relaxation oscillator (continued)

The folds are at $x = \pm a$, and the horizontal jump endpoints are $x = \mp 2a$.

Introduce the slow time

$$\tau = \frac{t}{\mu}.$$

In the singular reduced problem,

$$0 = y - F_0(x), \quad \frac{dy}{d\tau} = -x.$$

Consequently,

$$F'_0(x) \frac{dx}{d\tau} = -x, \quad \frac{d\tau}{dx} = -\frac{x^2 - a^2}{x}.$$

The right slow segment runs from $2a$ to a , so

$$\begin{aligned} T_{\text{right}}(\mu) &= \mu \int_a^{2a} \left(x - \frac{a^2}{x} \right) dx + o(\mu) \\ &= \mu \left[\frac{x^2}{2} - a^2 \ln x \right]_a^{2a} + o(\mu) \\ &= \mu a^2 \left(\frac{3}{2} - \ln 2 \right) + o(\mu). \end{aligned}$$

By symmetry, and because the fast transitions contribute $o(\mu)$,

$$\boxed{T(\mu) = \mu a^2 (3 - 2 \ln 2) + o(\mu)} \quad (\mu \rightarrow \infty).$$

Question: Audit a planar omega-limit conclusion

For a C^1 planar vector field, a positive semi-orbit lies in a compact subset of the vector field's domain, and its omega-limit set contains no equilibrium. What follows?

Answer. The Poincaré–Bendixson theorem makes the omega-limit set a periodic orbit.

Question: Audit an annulus conclusion

A compact annulus contains no equilibrium. Does a periodic orbit follow?

Answer. No. Positive invariance, or another form of forward confinement, must also be proved.

Question: Audit a first-integral conclusion

For a C^1 planar system, a connected compact regular level component of a first integral contains no equilibrium. What follows?

Answer. The level component is a periodic orbit. Nearby energy levels may form a family, so the orbit need not be a limit cycle.

Question: Audit a divergence conclusion

The divergence of a C^1 planar vector field is negative on an open, simply connected region containing a proposed closed orbit. What follows?

Answer. The Bendixson–Dulac criterion with $B \equiv 1$ rules out the proposed closed orbit.

Idea: Four layers of a complete qualitative analysis

state space and parameters $\Rightarrow$ equilibria and local data
 $\Rightarrow$ global invariant structure $\Rightarrow$ exact conclusion.